

Project Submission Package

Inter-Office Memo Management System

Student Name: Al Shabab
Student ID: 2523255630
Course Code: CSE226
Course Title: Foundations of Vibe Coding
Institution: North South University
Semester: Summer 2026
Submission Deadline: Midnight, 29 August 2026

1. Submission Links (As per Section 29)

REQUIREMENT	DESCRIPTION	SUBMISSION LINK
A. Deployed Application URL	Publicly accessible live deployment on Render (HTTPS)	https://inter-office-memo-system.onrender.com
B. Project Documentation	Comprehensive 26-section technical documentation & architecture report	<code>SYSTEM_DOCUMENTATION.md</code> / GitHub Documentation Link
C. Source Code Repository	Complete Git repository with commit history and setup instructions	https://github.com/shabab966/Project_03
C. Source Code ZIP URL	Direct download link to complete source code archive	https://github.com/shabab966/Project_03/archive/refs/heads/main.zip <i>(Local archive also generated on Desktop)</i>
D. AI Prompt & Response History	Chronological record of vibe-coding prompts, responses, refactorings, and fixes	<code>AI_PROMPT_RESPONSE_HISTORY.md</code> / GitHub AI History Link

2. Demonstration Accounts & Credentials (As per Section 23 & 29.E)

All accounts share the default password: `password123` (Or use the 1-Click "**Switch Persona (Demo)**" menu in the top navigation bar of the live site)

Organization 1: North South University (NSU)

NAME	EMAIL ADDRESS	ROLE	DEPARTMENT / DESIGNATION
Dr. M. Admin	admin@nsu.edu	ADMIN	Central Admin / Chief Information Officer
Prof. Dr. A. VC	vc@nsu.edu	USER	Executive Office / Vice Chancellor
Prof. Dr. Rajesh Palit	dean.seps@nsu.edu	USER	ECE Dept / Dean, SEPS
Dr. Chair ECE	chair.ece@nsu.edu	USER	ECE Dept / Department Chair
Mr. Finance Director	director.finance@nsu.edu	USER	Finance Office / Director of Finance
Dr. Alice Faculty	alice.faculty@nsu.edu	USER	CSE Dept / Associate Professor
Dr. Bob Faculty	bob.faculty@nsu.edu	USER	ECE Dept / Assistant Professor

Organization 2: Apex Global Tech (APEX) — Tenant Isolation Proof

NAME	EMAIL ADDRESS	ROLE	DESIGNATION
Sarah Admin	admin@apex.io	ADMIN	Head of Operations
David CEO	ceo@apex.io	USER	Chief Executive Officer
Elena VP	vp.eng@apex.io	USER	VP of Engineering
John Developer	john.doe@apex.io	USER	Staff Software Architect

Organization 3: Dhaka General Hospital (DGH) — Multi-Tenant Scalability Proof

NAME	EMAIL ADDRESS	ROLE	DESIGNATION
Dr. Robert Admin	admin@dgh.org	ADMIN	Medical Director & Chief Admin
Dr. Emily Watson	surgeon@dgh.org	USER	Chief of Surgery
Dr. Kevin Vance	icu@dgh.org	USER	Head of Critical Care Medicine

3. Step-by-Step Evaluator Demonstration Scenario (As per Section 28)

Follow these steps on the live application to evaluate all 14 mandatory demonstration milestones:

.og in as Author:

Go to <https://inter-office-memo-system.onrender.com/login>

Click the "Dr. Alice Faculty" demo card (or enter `alice.faculty@nsu.edu` / `password123`).

reate a Multi-Stage Memo:

Click "Create Office Memo" in the sidebar.

Title: `Procurement of High-Performance GPU Cluster for Deep Learning Lab` .

Category: `Procurement` | Priority: `Urgent` .

Workflow: Select "Procurement Request" template (Chair ECE $\rightarrow$ Dean SEPS $\rightarrow$ Finance Director $\rightarrow$ Vice Chancellor).

Click "Submit Office Memo".

Step 1 Review (Chair ECE):

Click **"Switch Persona (Demo)"** in the top navbar → Select **"Dr. Chair ECE"**.

Navigate to **"Inbox (Action)"** → Open the submitted memo.

Click **"Approve & Forward"** with comment: *"Approved from departmental budget perspective."*

Step 2 Change Request (Dean SEPS):

Switch persona to **"Prof. Dr. Rajesh Palit"** (Dean).

Open memo from Inbox → Click **"Request Changes"** with comment: *"Please specify exact model specifications (NVIDIA H100 vs A100)."*

Note the status changes to `CHANGES_REQUESTED` and returns to author.

Revision & Resubmission (Author):

Switch persona back to **"Dr. Alice Faculty"**.

Go to **"Sent Memos"** → Open the memo → Click **"Edit & Resubmit"**.

Update body text with model details → Enter change summary: *"Updated to 4x NVIDIA H100 SXM5."* → Click **"Resubmit Memo"**.

Notice that **Version 2** is created and the timeline preserves Version 1 history.

Final Approval Chain:

Switch to **Dr. Chair ECE** → Approve.

Switch to **Prof. Dr. Rajesh Palit** → Approve.

Switch to **Mr. Finance Director** → Approve.

Switch to **Prof. Dr. A. VC** → Final **Approve**.

Note that status is now `APPROVED` with official completion timestamp and read-only mode.

Delegation Demonstration:

Switch to **Prof. Dr. Rajesh Palit** → Navigate to **"Delegations"**.

Create a delegation granting authority to **"Dr. Bob Faculty"** for the next 7 days.

Switch to **Dr. Bob Faculty** → Note the navbar banner: *"Acting as Delegate for: Prof. Dr. Rajesh Palit"*.

Tenant Isolation Verification:

Switch persona to **"David CEO"** (Apex Global Tech).

Check Dashboard, Inbox, Sent, Search, and Audit Logs → Confirm **0 items** from North South University are visible.

4. Local Setup Instructions (As per Section 25)

```
`` bash
```

1. Clone repository

```
git clone https://github.com/shabab966/Project_03.git cd Project_03
```

2. Install dependencies

```
npm install
```

3. Setup environment variables

```
cp .env.example .env.local
```

4. Generate Prisma client & initialize database

```
npx prisma generate npx prisma db push node scripts/seed.js
```

5. Start development server

```
npm run dev
```

Open <http://localhost:3000>



5. The Vibe-Coding Process (As per Section 26.8)

5.1 AI Tools & Environment

The system was engineered using **Google Antigravity** (powered by Gemini 2.5 Pro and Claude 3.5 Sonnet) operating directly as an autonomous pair-programmer with terminal, filesystem, and code execution tools.

5.2 Requirement Communication & Step-by-Step Prompting

Development was executed through structured architectural prompts rather than monolithic generations:

Schema & Multi-Tenant Scoping: Designed Prisma data models with mandatory `organizationId` keys across all entities.

Deterministic State Machine: Engineered sequential step transitions ($A \rightarrow B \rightarrow C \rightarrow D$) with turn-based server validation.

Dual-Datasource Database Portability: Created `scripts/prepare-db.js` to automatically detect environment and switch between local SQLite (`dev.db`) and Render PostgreSQL (`DATABASE_URL`).

Email Authentication & Sandbox Fallback: Integrated Resend API with direct activation links for offline validation.

5.3 Error Diagnosis & AI-Assisted Debugging

Prisma Datasource Protocol Mismatch: Fixed runtime error where local SQLite received `postgresql://` validation by building an automatic pre-build datasource switcher.

Mobile Responsive Drawer Layout: Diagnosed viewport squishing on mobile screens and rebuilt `Sidebar.tsx` into a responsive slide-over drawer with backdrop blur.

Resend Sandbox 403 Delivery Constraint: Handled sandbox mode restrictions by adding 1-click direct activation and password-reset link cards in the UI.

5.4 Verification & Validation

Automated full-stack static page compilation (`npm run build`).

Automated database migrations and multi-tenant seeding (`node scripts/seed.js`).

Complete live end-to-end evaluation scenario walkthrough on Render.

6. Known Limitations & Technical Compromises (As per Section 26.9)

Email Delivery in Free Sandbox Mode: Outbound emails via Resend free tier are restricted to the registered developer address (tshabab26@gmail.com). To accommodate academic evaluation accounts (@nsu.edu), the UI plays instant copyable activation and password-reset links.

Ephemeral Disk Storage on Free Cloud Dynos: File attachments are stored locally under public/uploads/{orgId}/. On Render's free tier, local disk files reset when the dyno spins down after 15 minutes of inactivity (Recommended production upgrade: Amazon S3 / Cloudflare R2).

HTTP State Refresh vs. WebSockets: Notifications and inbox counts update via route navigation and API fetches rather than persistent WebSocket connections to preserve zero-cost serverless compatibility.

Single Active Proxy Delegation: A user can designate one active delegate per date window. Nested or multi-hop delegations ($A \rightarrow B \rightarrow C$) are restricted to prevent circular authorization loops.
